

Weitong Tian

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Summary

Electrical Engineering master's student with skills in **Python, MATLAB, Verilog, and Java**, with project experience spanning **LLM evaluation design, transfer-learning methods, and embedded computer vision**. Co-designed multi-model counterfactual-generation experiments using independent LLM judges, consensus voting, and semantic-consistency assessment. Designed MLP pretraining and fine-tuning experiments for circuit-performance prediction across operating conditions. Developing Raspberry Pi vision software using **perspective correction and lightweight image classification**. Additional experience includes ARM/RV32I processor design, pipeline hazard handling, cache architecture, and FPGA verification.

Education

University of Notre Dame

Sep. 2025 – Fall 2026 (expected)

South Bend, USA

- Master's student in Electrical Engineering.

Southern University of Science and Technology

Sep. 2021 – Jun. 2025

Shenzhen, China

- Bachelor of Engineering in Microelectronics | GPA: 3.79 | Class rank: 14/92.

University of California San Diego

Mar. – Sep. 1, 2024

San Diego, USA

- Exchange student.

Research Experience

GaN HEMT Analysis | Research Assistant

Sep. 2025 – Present

University of Notre Dame | Prof. Patrick Fay

- Characterize WIN Semiconductors and Northrop Grumman devices using **swept and pulsed I–V and S-parameter measurements**, C–V profiling, and transconductance overshoot analysis.
- Develop **HEMT models in ADS** and fit parameters to large-signal, small-signal, and I–V characteristics.
- Perform **load-pull measurements in the K, W, and D bands** and analyze device performance.

RFIC Design | Research Assistant

Jun. 2022 – Jun. 2025

Southern University of Science and Technology | Prof. Xiaoguang Liu

- Designed and taped out **3–10 GHz cryogenic low-noise amplifiers (LNAs)** using an **STMicroelectronics SiGe process**.
- Contributed to a resistance test-structure project for cryogenic LNAs through tape-out.
- Investigated algorithms for millimeter-wave radar.

Equal-P Antenna Research | Research Assistant

Apr. – Sep. 2024

University of California San Diego | Prof. Daniel Sievenpiper

- Participated in digitally tunable capacitor (DTC) design; soldered and measured existing DTC circuits.
- Implemented algorithms for automated antenna S-parameter measurements. Research summary.

Publications & Manuscripts

Ultra-Thin YAlN-Barrier GaN-Channel High Mobility Electron Transistors

Juncheng Xiong, Md Mehedi Hasan Tanim, Jiangnan Liu, Yu-En Jeng, Rui Shen, **Weitong Tian**, Wesley Turner, Pengcheng Xu, Weifeng Wu, Zetian Mi, and Patrick Fay.

Manuscript submitted, July 12, 2026.

Technical Skills

Programming & Digital Design: Python, MATLAB, Verilog, Java; Vivado; FPGA implementation and verification.

AI & Machine Learning: LLM evaluation, counterfactual text generation, multi-model judging and consensus voting; MLP surrogate modeling, transfer learning and fine-tuning; perspective correction, image classification, and Raspberry Pi computer vision (in progress).

Circuit & Device Tools: Cadence Virtuoso, Keysight ADS, Keysight IC-CAP, Ansys HFSS, Synopsys Sentaurus, Silvaco ATLAS, Lichuang EDA.

RF & Electrical Characterization: Swept/pulsed I–V and S-parameters, C–V profiling, W-band, K-band, and D-band load-pull measurements, transconductance analysis; TRL and LRRM calibration.

Fabrication Equipment: ALD, RIE (dry etching), photo/e-beam lithography, RTA, sputtering systems, and e-beam evaporators.

Selected Projects

AI & Machine Learning

Sequential LLM Counterfactual Evaluation | *LLMs* | *NLP* | *Evaluation Design*

- Co-designed an evaluation framework for six LLMs sequentially reversing movie-review sentiment, with four independent judges using majority voting.
- Defined label-recovery, judge-agreement, and semantic-similarity measures to distinguish sentiment consistency from accumulated information distortion.

Transfer Learning for RF IC Performance Prediction | *MLP Surrogates* | *Transfer Learning*

- Designed a transfer-learning study to adapt an LNA performance predictor across temperatures or process corners, predicting gain, noise figure, S11, and DC power.
- Designed comparisons of source-model reuse, fine-tuning, and training from scratch to assess transfer across operating conditions with limited simulation data.

Physical Go Board Recognition & Analysis — In Progress | *Raspberry Pi* | *Pi Camera* | *Computer Vision* | *Lightweight Machine Learning*

- Developing recognition of a physical 19×19 Go board using perspective correction and classification of stones and empty intersections. Planned extensions include move tracking and local KataGo analysis.

RF, Circuits & Devices

Antenna Design | *HFSS*

- Designed and simulated microstrip antennas and arrays, planar antennas on substrates, and broadband antennas to analyze electromagnetic performance.

FMCW Radar Modeling | *MATLAB*

- Developed a millimeter-wave FMCW radar model for multi-target detection, including algorithms to estimate target angle, velocity, and position.

IC Fabrication | *Cleanroom* | *Fabrication report*

- Completed the full **2- μ m CMOS fabrication flow on 100-mm silicon wafers**, from wafer preparation through device and circuit testing.
- Performed photolithography, oxidation and LOCOS isolation, implantation and activation, deposition, wet/dry etching, gate and contact formation, metallization, and passivation.
- Performed process metrology and characterized MOSFETs, CMOS inverters, and TLM structures; tested ring oscillators and sound chips.

Diode & CMOS Device Simulation | *Sentaurus* | *Silvaco ATLAS*

- Simulated diode and CMOS device structures.

Analog, Digital & PCB Design | *Cadence Virtuoso* | *Lichuang EDA*

- Designed a two-stage amplifier, implemented a multi-digit adder, and completed a board-level PCB design across coursework projects.

Programming & Digital Systems

Verilog Digital Design | *Verilog* | *Vivado* | *Processor design report*

- Implemented an arithmetic calculator and contributed to an ARM/RV32I processor project covering pipeline hazards, a four-way cache, floating-point arithmetic, and dual-issue execution, with simulation and FPGA board-test results.

Java Chess Project | *Java* | *GitHub*

- Developed a chess game with game logic and a graphical user interface.

Speech Denoising | *MATLAB*

- Reviewed artificial electronic cochlea research for a speech-denoising project.

Passive Radar Signal Processing | *MATLAB*

- Studied the ambiguity function and generated range-Doppler spectra.